

Md. Emran Biswas

📍 Dinajpur, Bangladesh ✉ emranbiswas.hstu@gmail.com 🏠 Google Scholar 🔗 LinkedIn

RESEARCH INTERESTS

I am enthusiastic about machine learning, optimization algorithms, and leveraging these technologies for societal impact. My research focuses on predictive analysis, optimization using machine learning models, bioinformatics-based drug discovery, and exploring applications that contribute to 'AI for Good.'

EDUCATION

- Hajee Mohammad Danesh Science & Technology University (HSTU) Dinajpur, Bangladesh
- BSc, Electronics and Communication Engineering Mar 2019 – Nov 2024

SKILLS

- **Programming Language:** C, C++, Java, Python
- **Query Language:** SQL, NoSQL (MongoDB)
- **Cloud:** AWS Cloud
- **Web Framework:** Django, Flask, Plotly-Dash, Streamlit
- **Deep Learning Framework:** Tensorflow, Pytorch
- **Scientific Computing Package:** Numpy, Scikit-learn, Pandas

EXPERIENCE

Research Assistant

Mar 2019 – Sep 2024

Hajee Mohammad Danesh Science and Technology University · Part-time

Lead: Dr. Md. Dulal Haque, MD SHAHZAMAL

- Researched machine learning algorithms, focusing on predictive modeling and optimization.
- Fine-tuned models and managed large datasets.
- Published two IEEE conference papers; two more accepted.

Research Assistant

Mar 2024 – Sep 2024

Hajee Mohammad Danesh Science and Technology University · Part-time

Lead: Dr. Tangina Sultana

- Developed ApexBoost Regression (ABR) model for demand forecasting.
- Created a flight arrival delay predictor app.
- Two journal paper under review in the International Journal of Data Science and Analytics

PUBLICATION

- Biswas, M.E., Shahzamal, M. and Haque, M.D., 2024, May. [Machine Learning Approach to Estimate Requirements for Target Productivity of Garments Employees](#). In 2024 6th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT) (pp. 921-926). IEEE.
- Isalm, Md & Biswas, & Haque, Dulal & Hossain, Md & Shahzamal, Md. (2023). [An Effective Data Driven Approach to Predict Bike Rental Demand](#).

ACCEPTED PAPERS

- An Adaptive Machine Learning Approach for Electrical Fault Detection and Minimization (ICCIT-2024, BUET)
- A Bioinformatics Approach to Identify Hub Genes Across Schizophrenia, Anxiety, Bipolar Disorder, and Depressive Disorder for Network-Based Drug Discovery (ICCIT- 2024, BUET)

UNDER REVIEW

- Spatio-Temporal Feature Engineering and Selection Based Flight Arrival Delay Prediction Using Deep Feedforward Regression Network (Electronics Journal)
- Spatio-Temporal Data-Driven Hourly Bike-Sharing Demand Prediction Using ApexBoost Regression (International Journal of Data Science and Analytics)

PROJECTS

Flight delay Prediction App | *Tensorflow, Scikit-learn, Pandas, Plotly Dash* May 2024 – Aug 2024

- We investigated approximately 3 million US airline data samples from 2019 to 2023.
- Achieved a 99.93 R-Square score in testing using feature engineering and our proposed deep learning model

Face Detection Based Attendance System | *Opencv, Flask, Tensorflow* Mar 2022 – May 2022

- We collected a comprehensive image dataset of students from our university to develop a model for facial recognition
- Trained CNN model to accurately predict and classify student identities, then deployed the model through a Flask application for real-time usage.

Diseases Detection System | *Scikit-learn, Pandas, Matplotlib, Flask* Sep 2022 – Dec 2022

- We trained a model to predict liver, pneumonia, kidney, diabetes, stroke, and heart diseases
- Deployed the model using Flask web framework.

AWARDS

First Runner-Up in Project Exhibition, 2022

Title: Face Detection-Based Automatic Attendance System

Second Runner-Up in Project Exhibition, 2023

Title: AI-Based Health Checking System

REFERENCES

- [Dr. Md. Dulal Haque](#), Professor, Department of Electronics and Communication Engineering, HSTU, Bangladesh, Phone: +8801712534968, Email: dhaque@hstu.ac.bd
- [MD SHAHZAMAL](#), PhD, Macquarie University, Australia, Email: mdszamal@gmail.com
- [Dr. Tangina Sultana](#), Post Doctoral Researcher, Kyung Hee University, South Korea, Associate Professor, Department of Electronics and Communication Engineering, HSTU, Bangladesh, Phone: +8801309964335, Email: dristy.043@gmail.com
- [Md. Selim Hossain](#), Lecturer, Department of Electronics and Communication Engineering, HSTU, Phone: +8801737598853, Email: selim.ece@tch.hstu.ac.bd